

Increasing Trust with Augmented Reality in Human-in-the-Loop Intelligent System: A Case Study with Inspection Task

Zhenning Zhang^{1,*} Haoyu Wang¹ Zhigeng Pan² Weiqing Li¹ Zhiyong Su¹

¹) Nanjing University of Science and Technology, China

²) Nanjing University of Information Science and Technology, China

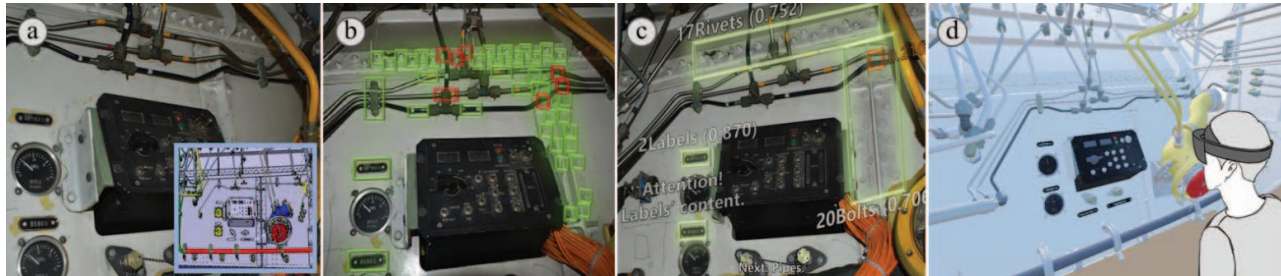


Figure 1. Illustration of three conditions: (a) production documents (PD); (b) with AR (AR); (c) AR with trust-increasing factors (ART). And (d) the AR-based inspection scenario with a digital model.

ABSTRACT

Augmented reality (AR) technologies combining computer vision algorithms have shown great potential to assist users in inspection tasks. However, as an intelligent system in which human-in-the-loop, the AR-based inspection system presents key challenges to human-machine trust. In this paper, we propose AR-based trust-increasing factors to enhance user trust in the inspection task. We explore the factors in a user study within an aircraft cabin inspection task that requires accurate inspection and strong confidence. The results insights help designers of AR applications provide a confident user experience in a human-in-the-loop intelligent system.

Index Terms: Human-centered computing—Interaction paradigms—Mixed/augmented reality;

1 INTRODUCTION

Augmented reality (AR) can overlay and display 3D digital content in the real world, convey expert knowledge to users, and promote the understanding of complex tasks. With the traditional methods (e.g. viewing the production documents) used in most inspection tasks, users had to mentally match the directions shown in the documents to directions in the real world. AR-based inspection systems can express the production documents and expert knowledge as virtual graphics, which are superimposed and aligned with the inspection scene on a real-world view. Users can visually compare the current production status with the graphics of the production document. On the other hand, computer vision algorithms provide a way of automatic inspection, which can examine the product hardware in real-time and show visual feedback to assist inspectors. Therefore, an AR-based inspection system can make the directions easier to perceive and reduce the cognitive load of inspectors theoretically.

Despite such huge potential, as an intelligent system in which human-in-the-loop assists inspectors, the AR-based inspection system will also present important challenges to human-machine

trust [1]. A lot remains to be done to let users confidently inspect with AR contents in inspection tasks.

Trust is particularly important to improve the partnerships between inspectors and our inspection system. Compared with other human-in-the-loop intelligent system, for example automated vehicle systems [2], users usually have driving knowledge, while users of our AR-based inspection cannot have the knowledge of production documents in advance. Therefore, trust is more important and difficult for our inspection system. Compared with the equipment maintenance assistance system or navigation system, their users can usually get actual feedback through real-world operations (e.g., completing the assembly of a part or moving to a specific location). Within our system, the inspectors can only rely on the AR content to inspect the product status and make decisions, and cannot obtain feedback information. While the concept of affective computing suggests that systems that can sense and respond to users' emotional states may greatly improve human-machine interaction. Therefore, it is particularly necessary to enhance users' trust in our inspection system, which will directly affect the acceptability and ease of use of the system.

2 AR CONTENT WITH TRUST-INCREASING FACTORS

We designed the trust-increasing factors with domain experts. We also got inspiration from the trust-increasing factors [2] of other human-in-the-loop intelligent systems, which include the transparency and uncertainty information for the cognitive forcing interventions [4], expert knowledge for the guidance of inspection, and anthropomorphic interaction for feedback as follows:

(1) Uncertainty Information. For simple results, without transparency, a significant fraction of users would get distracted and lost trying to understand exactly what happened [3]. Users would overreliance on the AR content of the results. The AR content should display the inspection confidence of each part which is calculated by the neural networks, instead of the result of detected or not.

(2) Expert Knowledge. A system perceived as “expert” feels more trustworthy. Except for the expert knowledge required by the system to realize the inspection function, experts' annotations and parts information are involved in the system as a trust-affecting factor. This expert information is authored offline, and it would be expressed as AR content while running the AR system.

* Corresponding author: ZhenningZhang@njust.edu.cn

(3) **Anthropomorphic Interaction.** To avoid employing the advantages of AR HMD (e.g., hands-free), eye gaze is adopted as an interactive mode for the system to push information to the inspector. The place where the eyes stay or the time they stay will bring clues to the information pushed by the system. It can be a trustworthy way to convey feedback or information.

3 USER STUDY

To explore the benefits and limitations of the AR-based inspection system with trust-increasing factors, we evaluated it in a user study. We recruited 16 participants (2 females and 14 males) from the college and aircraft research institute, to participate in this study.

The goal of the study was to test and evaluate the cognitive load and trust in using the AR-based inspection system, by comparing the results across the three conditions of inspection including inspection with production documents (PD), with AR visualization (AR), and with AR and trust-increasing factors (ART), shown in Figure 1.

Cognitive Load. Figure. 2 shows plot for the cognitive load in NASA-RTLX questionnaire subscales.

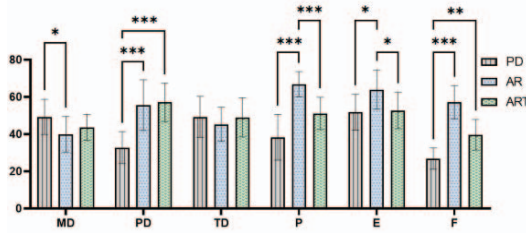


Figure. 2. Cognitive load by NASA-RTLX questionnaire.

Trust. Figure. 3 shows plot for the McKnight Trust questionnaire subscales reliability, functionality, helpfulness, situational normality, faith in general technology, and trusting stance. The questionnaire had a high level of reliability, with a Cronbach's α value of 0.760.

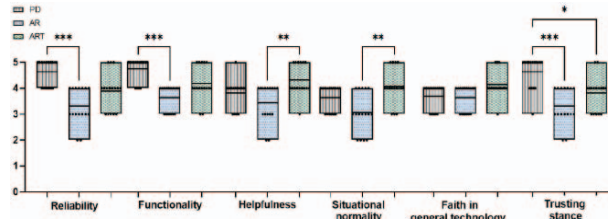


Fig. 3. Trust evaluation results by McKnight Trust questionnaire.

User Feedback. Here we report the comments received from the participants during the interview. We sorted the comments, divided these comments into two parts (cognitive load, and user trust), and selected representative comments for analysis.

(1) Cognitive load. In addition to the Physical Demand brought by the comfort mentioned in the usability of the system, participants also gave opinions on the Mental Demand.

"Compared with the condition with the production documents, I think I will rely on the results displayed by AR and lack my thinking." "I think the AR with trust affecting factors can interact with me and generate interactive feedback, and make me more satisfied." "The system provides a visual measurement function, which greatly facilitates our inspection task."

Participants surmised that since the raw AR-based system is only one-way output (the detected content is displayed through AR) which lacks the loop of interaction and feedback, makes them uncomfortable during the task. However, they also pointed out that trust in the AR-based inspection system affects the workload of the task, and bad trust will bring high frustration during the inspection task.

(2) Trust. Trust in the human-in-the-loop intelligent system is the most important argument in this paper, so we also received some positive comments about the trust in our AR-based inspection system.

"I found that AR content with uncertain information and expert knowledge can make me feel more confident than the AR visualization with no factor." "If only the inspection content of AR is displayed, it may cause me to rely too much on it, and the other content provided by AR gives me a chance to think and work with an intelligent system." "I think this technology is trustworthy, but compared with traditional methods with production documents, I personally think there is still a gap. Maybe users need some time to adapt to it."

4 FUTURE WORK.

Future, the AR-based inspection system could integrate multimodal interaction and visualization methods with trust-increasing factors. An exciting solution includes the attachment of voice prompting and online expert collaboration may be able to further improve user trust. We make this conjecture due to the online social behaviour of experts having the potential to improve the psychological needs and emotional needs of the individual which may be beneficial to the trust with AR.

5 CONCLUSION

We present a case study in inspection task with trust-increasing factors. In our user study, users obtained a higher cognitive load and a lower trust in the AR-based method than the traditional method in the inspection task. We verified that trust would affect the acceptability and ease of use of the system, thus affecting the psychological needs of users in the inspection task. Moreover, the trust-increasing factors we proposed with cognitive forcing interventions, knowledge, and interaction, enhanced the trust of users compared with the raw AR content significantly.

We believe that AR combined with artificial intelligence will become an important visualization technology for the assistance of visual perception in the future. Design challenges still exist regarding how to improve the trust of users in AR visualization. The AR-based inspection system with trust-increasing factors is an important step towards designing confident AR visualization for the human-in-the-loop intelligent system.

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REFERENCES

- [1] Honeycutt, Donald, Mahsan Nourani, and Eric Ragan. "Soliciting human-in-the-loop user feedback for interactive machine learning reduces user trust and impressions of model accuracy." Proceedings of the AAAI Conference on Human Computation and Crowdsourcing. Vol. 8. 2020.
- [2] Ekman F, Johansson M, Sochor J. "Creating appropriate trust in automated vehicle systems: A framework for HMI design." IEEE Transactions on Human-Machine Systems, 2017, 48(1): 95-101.
- [3] Dasgupta, Aritra, et al. "Familiarity vs trust: A comparative study of domain scientists' trust in visual analytics and conventional analysis methods." IEEE transactions on visualization and computer graphics 23.1 (2016): 271-280.
- [4] Bucinca Z, Malaya M B, Gajos K Z. "To trust or to think: cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making." Proceedings of the ACM on Human-Computer Interaction, 2021, 5(CSCW1): 1-21.